

Implicit Manifold Entanglement:

Resolving Low-Dimensional Stochasticity via Koopman-Regularized Conditional Flow Matching

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Abstract

Physical reality is often modeled as a closed system, yet observed trajectories frequently exhibit stochastic bifurcations that defy low-dimensional deterministic explanation. We posit that these systems are low-rank projections of a higher-dimensional, infinite manifold where the dynamics remain deterministic. In this proposal, we introduce **Manifold-Coupled Flow Matching (MC-FM)**, a novel framework that resolves aleatoric uncertainty by explicitly modeling unobserved dimensions as a *Koopman Invariant Subspace*. We leverage Takens’ Embedding Theorem to reconstruct the latent topology and employ Neural Stochastic Differential Equations (SDEs) to model the coupling. Crucially, we propose a rigorous topological validation method using Persistent Homology (Betti numbers) to prove that the learned latent structure is isomorphic to the ground-truth manifold, distinguishing this work from standard generative modeling.

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1 Introduction and Motivation

1.1 The Metaphysical Intuition: The “Soul” of the System

In classical philosophy and metaphysics, the physical body is often viewed as a projection of a deeper, non-physical entity (the “Soul” or “Spirit”) that influences the body’s trajectory. Under this view, moments that appear to be random physical bifurcations (choices) are actually deterministic outcomes driven by the state of the unobserved dimension.

While this is a philosophical abstraction, it maps directly to a fundamental problem in Control Theory and Physics: **Partial Observability of Chaotic Systems**. If we treat the “Soul” not as a mystical entity, but as a set of *latent, slow-moving variables* governed by their own internal logic, we can resolve the apparent randomness of the physical world.

1.2 The Scientific Realization

In this research, we formalize this intuition using rigorous terminology. We utilize the following mapping to translate the metaphysical concept into a solvable engineering problem:

Table 1: Terminological Mapping for Research Formulation

Metaphysical Concept	Academic/Dynamical Systems Terminology
The Physical Body	The Observable State Space ($x_t \in \mathbb{R}^d$)
The Spiritual/Hidden Dimension	The Latent Manifold / Invariant Subspace ($y_t \in \mathbb{R}^k$)
The “Ghost” Influence	Non-Zero Coupling Terms (Schur Complement)
Divine Order / Truth	The Koopman Operator (Global Linearization)

Our hypothesis is that by modeling the “Hidden Dimension” (y_t) as a globally linear, stable system (via Koopman theory), we can stabilize the prediction of the “Physical Body” (x_t) even in chaotic regimes.

2 Mathematical Problem Formulation

2.1 The Universal System (SDE Formulation)

Let the universal state z_t evolve on a smooth manifold $\mathcal{M} \subseteq \mathbb{R}^D$. We decompose the state into observable (x) and latent (y) components. The evolution is governed by a Coupled Stochastic Differential Equation (SDE):

$$d \begin{bmatrix} x_t \\ y_t \end{bmatrix} = \underbrace{\begin{bmatrix} f(x_t, y_t) \\ g(y_t, x_t) \end{bmatrix} dt}_{\text{Coupled Drift}} + \underbrace{\begin{bmatrix} \sigma_x dW_t^x \\ \sigma_y dW_t^y \end{bmatrix}}_{\text{Diffusion}} \quad (1)$$

Where:

- x_t represents the observable trajectory (e.g., position, price, temperature).
- y_t represents the latent manifold (e.g., hidden forces, market sentiment, deep currents).
- $\frac{\partial f}{\partial y} \neq 0$ implies the latent state actively forces the physical state.

The Challenge: We only possess discrete observations of x_t . The inverse mapping $x_t \rightarrow y_t$ is ill-posed because multiple latent states can map to the same physical coordinate (the bifurcation problem).

3 Theoretical Anchors

To validate the reconstruction of the unobserved manifold, we rely on a triad of foundational theorems:

3.1 I. Takens' Embedding Theorem (Geometry)

Theorem: The topology of a multivariate dynamical system can be reconstructed from the time-delay embedding of a single observable variable.

$$\Phi(x) = [x_t, x_{t-\tau}, x_{t-2\tau}, \dots] \quad (2)$$

Application: This justifies our use of a Sequence Encoder (Transformer) on the history of x to infer the current state of y . It proves that the information of the “Soul” is encoded in the history of the “Body.”

3.2 II. Koopman Operator Theory (Global Stability)

Theorem: There exists an infinite-dimensional linear operator $\mathcal{K}$ that advances all observables g of the state space: $\mathcal{K}g(z_t) = g(z_{t+1})$. *Application:* We assume the latent manifold y_t lies on the subspace spanned by the dominant eigenfunctions of $\mathcal{K}$. This imposes a **Linearization Constraint** on the latent space, ensuring y_t is not just random noise, but a coherent, slowly evolving structure.

3.3 III. Schur Complement (Coupling Strength)

Theorem: In a block-partitioned system matrix, the stability of the visible block is determined by the feedback term $BD^{-1}C$. *Application:* By learning the coupling functions, we effectively estimate this Schur complement, mathematically quantifying the “influence” of the hidden dimensions on physical stability.

4 Methodology: MC-FM Architecture

We propose the **Manifold-Coupled Flow Matcher (MC-FM)**. The architecture consists of three differentiable modules trained end-to-end.

4.1 Module A: The Koopman Encoder (Latent Reconstruction)

- **Input:** Sliding window history $X_{hist} \in \mathbb{R}^{T \times d}$.
- **Model:** Transformer with Multi-Head Attention.
- **Output:** Latent state estimate $\hat{y}_t$.

- **Constraint:** We impose a Koopman Loss to enforce linear evolution in the latent space:

$$\mathcal{L}_{Koopman} = \sum_t \|\hat{y}_{t+1} - \mathbf{K}\hat{y}_t\|^2 \quad (3)$$

where $\mathbf{K}$ is a learnable linear matrix approximating the Koopman operator.

4.2 Module B: The Coupled Score Network (Flow Matching)

Function: Approximates the time-dependent vector field v_t that pushes a noise distribution towards the data distribution, conditioned on the latent manifold.

$$\mathcal{L}_{Flow} = \mathbb{E}_{t,x,\hat{y}} [\|v_\theta(x_t, \hat{y}_t, t) - u_t(x|x_1)\|^2] \quad (4)$$

This network (U-Net or ResNet MLP) takes the physical state and the inferred latent state to predict the gradient of the flow.

4.3 Module C: SDE Integrator

We utilize a differentiable solver (Euler-Maruyama) with the **Adjoint Sensitivity Method** to solve the Initial Value Problem (IVP) forward in time. This allows for memory-efficient backpropagation through the differential equation dynamics.

4.4 Optimization

The model is trained via a composite loss function balancing generative fidelity and spectral consistency:

$$\mathcal{L}_{Total} = \mathcal{L}_{Flow} + \lambda \mathcal{L}_{Koopman} \quad (5)$$

5 Experimental Plan

5.1 Phase 1: The “Universe” Generation (Lorenz-96)

We will simulate the high-dimensional Lorenz-96 system ($D = 10$) and mask 8 dimensions, treating them as the hidden manifold. We will explicitly identify bifurcation points where the 2D projection intersects itself.

- **Goal:** Establish a ground truth dataset $\mathcal{D} = \{(x_t, y_{true})\}$ to verify if the model can recover y_{true} solely from x .

5.2 Phase 2: Topological Validation (The Differentiator)

This is the critical validation step. We will not just compare Mean Squared Error (MSE), but the **Topological Signature** of the learned manifold.

- **Method:** We will compute the **Persistent Homology** of the inferred latent space $\hat{y}$ and the true hidden space y_{true} .
- **Metric:** Comparison of **Betti Numbers** (β_0 connected components, β_1 loops/voids) and the Wasserstein distance between Persistence Diagrams.
- **Hypothesis:** If the model has truly learned the “Spirit” of the system (and not just memorized noise), then $\beta_n(\hat{y}) \approx \beta_n(y_{true})$.

5.3 Phase 3: The “Ghost” Bifurcation Experiment

We will visualize the influence of the latent variable at a bifurcation point.

- **Step 1:** Select a crossing point x_{cross} where the physical path is ambiguous.
- **Step 2:** Initialize model with x_{cross} and Latent State A (y_a). $\rightarrow$ Predict Path A.
- **Step 3:** Initialize model with x_{cross} and Latent State B (y_b). $\rightarrow$ Predict Path B.
- **Showcase:** A 2D phase-space plot showing two distinct futures emerging from the *exact same* physical coordinate, controlled solely by the spiritual coordinate.

5.4 Phase 4: Robustness to Noise

We will inject Gaussian noise into the physical input x . We hypothesize that the Koopman-regularized latent state will act as an “anchor,” reducing the drift variance compared to standard Neural SDEs.

5.5 Phase 5: Real-World Validation (El Niño / ENSO)

We will apply the model to the NOAA El Niño Southern Oscillation dataset.

- **Observable (x):** Sea Surface Temperature (SST) in the Pacific (Visible).
- **Latent Target (y):** Deep ocean thermocline depth (Hidden/Masked).
- **Goal:** Predict the onset of El Niño events (bifurcations from neutral to warming) earlier than standard autoregressive models by inferring the hidden energy buildup in the “Latent” deep ocean.

6 Timeline

Table 2: Execution Roadmap

Phase	Duration	Key Deliverables
1. Data Gen	Weeks 1-2	Lorenz-96 Bifurcation Dataset
2. Topology	Weeks 3-4	Implementation of Koopman Encoder & TDA (Betti Plots)
3. Core Model	Weeks 5-7	MC-FM Training (Flow Matching + Koopman Loss)
4. Experiments	Weeks 8-9	“Ghost” Bifurcation Plots & Real-World ENSO Validation
5. Writing	Week 10	Final Thesis/Paper Compilation

7 Conclusion

This research moves beyond standard generative modeling by treating latent variables as **physically real, topologically consistent manifolds**. By integrating **Koopman Operator Theory** with **Stochastic Flow Matching**, we provide a framework that mathematically validates the concept of multi-dimensional co-existence. We aim to demonstrate that what appears as stochasticity in the physical world is often geometry in the high-dimensional world, offering a new paradigm for predicting complex, partially observed systems.

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